



Explore

Summary Chart of Naming Compounds

Ionic Compounds



2 Elements: Metal + Nonmetal

Name metal first, then change the nonmetal ending to “-ide.”

NaCl = Sodium chloride

For elements with multiple charges (transition metals), use Roman numeral.

FeO = Iron (II) chloride

Fe₂O₃ = Iron (III) chloride

More than 2 elements: cation + anion

Name metal first (or the cation), then the nonmetal (or anion).

KNO₃ = Potassium nitrate

NaOH = Sodium hydroxide

See polyatomic ion chart. Most are anions, except for NH₄⁺. If ammonium is used, then the anion ends in “-ide.”

NH₄Cl = ammonium chloride

Polyatomic Ions

Acetate C₂H₃O₂⁻

Ammonium NH₄⁺

Carbonate CO₃²⁻

Chlorate ClO₃⁻

Chlorite ClO₂⁻

Chromate CrO₄²⁻

Cyanide

CN⁻Dichromate

Cr₂O₇²⁻Hydrogen

carbonate HCO₃⁻

Naming Acids



H + non-oxyacid (anion)

Use the prefix **hydro-** and change the anion ending to “-ic.”

HCl = Hydrochloric acid

H + oxyacid (anion)

Change the ending of the oxyacid (anion). If it is “-ate,” the ending becomes “-ic.” If it is “-ite,” the ending becomes “-ous.”

H₂SO₄ = Sulfuric acid

H₂SO₃ = Sulfurous acid

Hydroxide OH⁻

Hypochlorite

HClO₂Nitrate

NO₃⁻Nitrite

NO₂⁻Perchlorate

ClO₄⁻Permanganate

MnO₄⁻Phosphate

PO₄³⁻Sulfate

SO₄²⁻Sulfite

SO₃²⁻

Covalent Compounds



2 Elements: Nonmetal + Nonmetal

Use a prefix for each element in the compound (count the atoms in the formula). Name the first element, and change the second element ending to “-ide.”

N₂O₅ =

Dinitrogen pentoxide

Prefixes

- 1 *mono-*
- 2 *di-*
- 3 *tri-*
- 4 *tetra-*
- 5 *penta-*
- 6 *hexa-*
- 7 *hepta-*
- 8 *octa-*
- 9 *nona-*
- 10 *deca-*